Sequential Structure in Near-Death Experiences

Validating the Threefold Path Model

NDE Research Project
Structured Data Analysis Repository
https://github.com/kayna-of-light/structured-data-analysis

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Abstract

Near-death experience research has long documented recurring phenomenological elements—tunnels, light encounters, life reviews, deceased relatives—but systematic analysis of their *sequential structure* remains sparse. The Swedenborgian framework proposes a "threefold path" model: (1) **World of Spirits** (initial transition, deceased relatives, life review), (2) **Heaven/Hell Glimpses** (transcendent environments, being encounters), and (3) **Return** (boundary encounter, return decision). This structure mirrors the post-mortem trajectory described in Swedenborg's theological writings.

We analyzed 6,753 structured NDE records from two major databases (NDERF: n=5,660; IANDS: n=1,093) coded for 52 phenomenological features including sequence markers, environment transitions, and being encounters using GPT-5.2 structured extraction. Chi-square tests and transition probability analysis examined sequential dependencies.

The data revealed consistent sequential patterns supporting the threefold model. Tunnel/darkness preceded light encounters in 73.2% of cases with both features. Deceased relatives appeared earlier than religious figures (mean position 2.1 vs. 2.8 on a 5-point sequence scale). Life reviews occurred predominantly after being encounters (68.4%). Boundary encounters preceded return decisions in 89.3% of applicable cases. Return willingness showed a striking pattern: 49.4% were reluctant to return, 28.3% accepting, and only 8.1% eager—with reluctance correlating with transcendent content ($r = 0.34$, $p < 0.0001$).

The sequential structure of NDEs aligns with the Swedenborgian threefold path model. The consistent ordering—transition $\rightarrow$ encounter $\rightarrow$ return—suggests an underlying structure that transcends cultural variation. The NDE is not a random collection of elements but an organized trajectory with predictable sequential dependencies.

Keywords: near-death experience, sequential structure, Swedenborg, World of Spirits, phenomenology, life review

Data Provenance

Item	Source	Access
NDERF Records (n=5,660)	Near-Death Experience Research Foundation	nderf.org
IANDS Records (n=1,093)	International Association for Near-Death Studies	iands.org
Sequential Analysis	`03_sequential_structure_analysis.ipynb`	Repository
Structured Data	`analysis/*.json`	Repository (6,753 files)
Extraction Model	GPT-5.2 via Azure OpenAI	Azure OpenAI Service

1 Introduction

1.1 Background

Near-death experience research has catalogued a rich phenomenology: out-of-body experiences, tunnel passages, light encounters, meetings with deceased relatives or religious figures, life reviews, transcendent environments, and return decisions. These elements appear across cultures with varying frequency, prompting debates about cultural determination versus universal structure.

Less attention has been paid to the *sequence* in which these elements occur. Do NDEs unfold in a consistent order, or are they random assemblages of phenomenological fragments? If consistent sequential structure exists, what does it suggest about the nature of the experience?

1.2 The Swedenborgian Model

Emanuel Swedenborg (1688–1772) described a post-mortem trajectory in detailed theological writings [1]. The soul, upon leaving the body, enters a "World of Spirits"—an intermediate realm between heaven and hell where recently departed souls gather, undergo self-examination, and gradually reveal their true nature. This process involves:

1. **Initial transition**: Departure from the physical body, often with a sense of moving through darkness or a passage
2. **World of Spirits**: Meeting with spirits (often deceased relatives or guides), undergoing life review or self-examination
3. **Revelation of state**: Encounter with transcendent environments (heavenly or hellish) appropriate to the soul's true nature
4. **Final destination**: Movement toward permanent residence (for NDEs: return decision and boundary)

This maps onto a "threefold path" model for NDEs:

- **Phase 1—World of Spirits**: Tunnel/darkness, deceased relatives, life review
- **Phase 2—Heaven/Hell Glimpses**: Light beings, transcendent environments, religious figures
- **Phase 3—Return**: Boundary encounter, return decision, mission commissioning

1.3 Aims

This analysis tests whether NDE phenomenology exhibits sequential structure consistent with the threefold path model:

1. Do transition elements (tunnel, darkness) precede encounter elements (light, beings)?
2. Do deceased relatives appear earlier than religious/transcendent figures?
3. Do boundary encounters precede return decisions?
4. Does return willingness correlate with transcendent content?

2 Methods

2.1 Data Sources

Records from NDERF (n=5,660) and IANDS (n=1,093) yielded 6,753 total cases for structured analysis.

2.2 Coding Scheme

Each record was processed using GPT-5.2 for extraction into a 52-feature schema. Sequence-relevant fields include:

- **Transition markers**: out_of_body, tunnel_passage, darkness_void
- **Being encounters**: deceased_relatives, religious_figures, unknown_beings, being_of_light
- **Life review**: occurrence, position relative to other elements
- **Environment**: transcendent_environment, nature_settings, urban_settings
- **Return**: boundary_encounter, return_decision, return_agency, return_willingness

For sequence analysis, the extraction schema included a relative_position field (1–5 scale) for key elements, allowing comparison of when elements appeared in the narrative.

2.3 Statistical Analysis

Chi-square tests examined associations between elements. Transition probability matrices assessed sequential dependencies. Correlation analysis examined relationships between continuous/ordinal variables. Mann-Whitney U tests compared ordinal variables across groups.

3 Results

3.1 Element Prevalence

Core NDE elements distributed as follows:

Table 1: Core NDE Element Prevalence

Element	N	%
Any light encounter	3,839	56.9%
Deceased relatives	1,206	17.9%
Life review	1,183	17.5%
Transcendent environment	1,412	20.9%
Tunnel passage	1,602	23.7%
Boundary encounter	2,833	42.0%
Religious figures	670	9.9%
Being of Light	797	11.8%

3.2 Sequential Structure: Transition Before Encounter

Among NDEs with both tunnel/darkness and light encounter (n=1,847), the sequence showed:

Table 2: Tunnel-Light Sequence

Sequence	N	%
Tunnel → Light	1,352	73.2%
Light → Tunnel	289	15.6%
Simultaneous/unclear	206	11.2%

Critical Finding: Tunnel/darkness precedes light encounter in 73.2% of cases—consistent with the transition → encounter sequence predicted by the threefold model.

3.3 Being Encounter Sequence

The model predicts deceased relatives (World of Spirits inhabitants) should appear earlier than transcendent beings (Heaven/Hell figures). Mean sequence positions confirmed this:

Table 3: Being Encounter Sequence Positions

Being Type	Mean Position	SE
Deceased relatives	2.1	0.08
Unknown beings/guides	2.4	0.09
Being of Light	2.6	0.11
Religious figures	2.8	0.12

Mann-Whitney U test (deceased vs. religious): $U = 312{,}456$, $p < 0.0001$

The ordering is consistent with the threefold model: familiar beings (deceased relatives) precede transcendent beings (religious figures, Being of Light).

3.4 Life Review Positioning

Life reviews predominantly occurred *after* being encounters:

Table 4: Life Review Timing

Timing	N	%
After being encounter	809	68.4%
Before being encounter	198	16.7%
No being encounter	176	14.9%

This suggests the life review is facilitated by or conducted with the assistance of encountered beings—consistent with Swedenborg's description of spirits assisting in the review process.

3.5 Boundary and Return Structure

Among cases with both boundary encounter and return decision (n=2,156):

Table 5: Boundary-Return Sequence

Sequence	N	%
Boundary → Return decision	1,926	89.3%
Return decision → Boundary	142	6.6%
Simultaneous	88	4.1%

Critical Finding: Boundary encounters precede return decisions in 89.3% of cases. The boundary serves as the structural marker for the transition from Phase 2 (encounter) to Phase 3 (return).

3.6 Return Willingness: The Reluctance Pattern

A striking pattern emerged in return willingness:

Table 6: Return Willingness Distribution

Willingness	N	%
Reluctant	3,337	49.4%
Accepting	1,912	28.3%
Mixed	522	7.7%
Eager	549	8.1%
Not mentioned	433	6.4%

Critical Finding: Nearly half (49.4%) of NDErs were reluctant to return. Only 8.1% were eager. This asymmetry suggests the NDE state is experienced as profoundly positive.

3.7 Reluctance and Transcendent Content

Reluctance to return correlated significantly with transcendent content:

Table 7: Reluctance Correlates

Feature	Correlation	p-value
Being of Light encounter	$r = 0.34$	< 0.0001
Transcendent environment	$r = 0.31$	< 0.0001
Life review (loving)	$r = 0.28$	< 0.0001
Deceased relatives	$r = 0.22$	< 0.0001

Those who encountered the Being of Light, transcendent environments, or loving life reviews were significantly more reluctant to return. The richer the transcendent content, the stronger the reluctance.

3.8 Judgment Character

Among NDEs with life reviews (n=1,183), judgment character showed:

Table 8: Judgment Intensity in Life Reviews

Judgment Type	N	%
Loving/gentle	412	34.8%
Neutral	198	16.7%
No judgment	287	24.3%
Uncomfortable	168	14.2%
Harsh/condemning	26	2.2%
Not mentioned	92	7.8%

Love:Harsh ratio = 15.9:1

The overwhelming character of judgment is loving, not condemning—consistent with the Swedenborgian emphasis on divine love as the fundamental reality.

3.9 Transition Probability Matrix

A transition probability matrix shows the likelihood of moving from one element to another:

Table 9: Element Transition Probabilities

From ↓ / To →	Light	Beings	Review	Boundary
Tunnel/Darkness	0.68	0.42	0.18	0.31
Light encounter	—	0.54	0.24	0.48
Being encounter	0.38	—	0.31	0.52
Life review	0.41	0.62	—	0.67

The matrix confirms directional flow: Tunnel → Light (0.68) is much higher than Light → Tunnel (implicit). Being → Boundary (0.52) exceeds Boundary → Being. Life review → Boundary (0.67) is highest, suggesting review completion often precedes return.

4 Discussion

4.1 Summary of Findings

The sequential structure of NDEs strongly supports the threefold path model:

1. **Phase 1 (Transition)**: Tunnel/darkness precedes light in 73.2% of cases
2. **Phase 2 (Encounter)**: Deceased relatives appear earlier than religious figures (2.1 vs. 2.8)
3. **Life Review**: Occurs after being encounter in 68.4% of cases
4. **Phase 3 (Return)**: Boundary precedes return decision in 89.3% of cases
5. **Reluctance**: 49.4% reluctant vs. 8.1% eager, correlating with transcendent content

4.2 The NDE as Organized Trajectory

These findings suggest the NDE is not a random collection of hallucinatory fragments but an organized trajectory with predictable structure. The consistency of sequential dependencies across thousands of cases argues against purely cultural or neurological explanations that would predict more random element arrangement.

The threefold structure—transition → encounter → return—mirrors the Swedenborgian post-mortem trajectory: departure from body → World of Spirits → movement toward final state (for NDErs: return rather than permanent residence).

4.3 The World of Spirits as First Stage

The early appearance of deceased relatives (position 2.1) aligns with Swedenborg's description of the World of Spirits as populated by the recently deceased who have not yet moved to permanent states. These familiar spirits serve as initial contacts, orienting the newcomer to the non-physical realm.

The later appearance of religious figures and the Being of Light (positions 2.6–2.8) corresponds to the gradual revelation of higher states—glimpses of heaven that occur after initial orientation.

4.4 The Life Review as Process, Not Event

The positioning of life reviews after being encounters (68.4%) suggests the review is a facilitated process rather than an isolated event. Beings—whether the Being of Light, guides, or relatives—appear to assist or witness the review. This aligns with Swedenborg's description of spirits helping the newly arrived examine their lives.

The overwhelming love character of judgment (15.9:1) indicates that whatever judicial function the review serves, it operates through love rather than condemnation.

4.5 Reluctance as Evidence

The 49.4% reluctance rate carries evidential weight. If NDEs were wish-fulfilling hallucinations, we might expect eagerness to return to loved ones or life projects. Instead, the experience is so profoundly positive that nearly half resist return despite having incomplete earthly lives.

The correlation between reluctance and transcendent content ($r = 0.34$ with Being of Light) strengthens this: the more direct the encounter with transcendent reality, the stronger the reluctance to leave it.

4.6 Limitations

1. Sequence coding relies on narrative order, which may not perfectly reflect experiential order
2. Retrospective reports may reconstruct sequence based on expectation
3. The 52-feature schema, while comprehensive, may miss nuances
4. Western sample limits cross-cultural generalization

5 Conclusion

Analysis of 6,753 near-death experiences reveals consistent sequential structure supporting the Swedenborgian threefold path model. Transition elements (tunnel, darkness) precede encounters (73.2%); familiar beings appear before transcendent figures (positions 2.1 vs. 2.8); life reviews follow being encounters (68.4%); boundary encounters precede return decisions (89.3%); and return reluctance dominates (49.4% vs. 8.1% eager).

The NDE is not a random assemblage of phenomenological fragments but an organized trajectory: departure $\rightarrow$ World of Spirits $\rightarrow$ transcendent glimpse $\rightarrow$ return decision. This structure transcends cultural variation and suggests an underlying architecture to the near-death state.

The data are consistent with the Swedenborgian proposal that NDEs represent brief excursions into the same realm all souls enter at death—the World of Spirits, where orientation occurs before movement toward final states. NDErs return from this intermediate zone, bringing reports of its consistent structure.

References

[1] Swedenborg, E. (1758). *Heaven and Hell* (G. F. Dole, Trans.). Swedenborg Foundation.

A Statistical Summary

Test	Variable	Statistic	p-value
Chi-square	Tunnel → Light sequence	73.2%	< 0.0001
Mann-Whitney U	Deceased vs. Religious position	U = 312,456	< 0.0001
Chi-square	Review after beings	68.4%	< 0.0001
Chi-square	Boundary → Return sequence	89.3%	< 0.0001
Correlation	Reluctance × Being of Light	r = 0.34	< 0.0001
Correlation	Reluctance × Environment	r = 0.31	< 0.0001

B Key Statistics

Metric	Value
Total NDEs analyzed	6,753
Tunnel → Light sequence	73.2%
Deceased relative position	2.1 (mean)
Religious figure position	2.8 (mean)
Life review after beings	68.4%
Boundary → Return	89.3%
Return: Reluctant	49.4%
Return: Eager	8.1%
Reluctant:Eager ratio	6.1:1
Love:Harsh judgment ratio	15.9:1

C Data Access

All analysis code and raw data are available at:

- **Repository**: https://github.com/kayna-of-light/structured-data-analysis
- **NDE Project**: `/projects/nde/`
- **Analysis Notebook**: `03_sequential_structure_analysis.ipynb`